Roll Number:_____

Thapar Institute of Engineering & Technology, Patiala
Department of Electrical and Instrumentation Engineering
Auxiliary Examination

B.E.	Course Code: UBM605 Course Name: Data Structures and Algorithms
12th August 2024	Monday, 5:30 – 8:30 PM
Time: 3 Hours, M. Marks: 100	Name Of Faculty: Dr. (Ms.) Mukesh Dalal

Note: All five questions are compulsory.

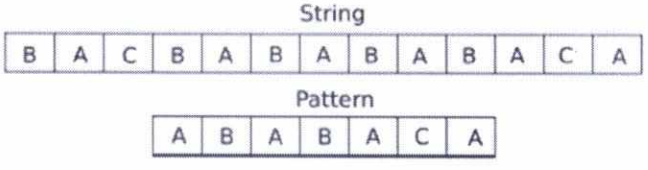
Q.1 a) Compare and contrast with the help of a suitable example for each (12)
(i) Array and Linked List
(ii) Binary and Linear search
(iii) Big-O and Big-Ω notations

b) State and solve the time complexity recurrence relation for each one of the following (8)
(i) Binary Search
(ii) Recursive solution to Tower of Hanoi problem

Q.2 a) Write algorithms/pseudocode for the basic operations of a stack using array or linked list. (8)

b) What is hashing? Explain the types of collision resolution techniques in hashing with suitable examples. (12)

Q.3 a) Consider the instance of the string matching problem given in the figure below: (8)



(i) Compute the LPS array for the Pattern given in the above figure.

(ii) Illustrate the working of KMP algorithm by applying it on the given instance of string matching problem.

b) Sort the following list using Insertion sort. Show each pass of the sort and discuss its run time complexity 78, 23, 56, 33, 59, 99, 12, 10, 90, 75 (12)

Q.4 a) What is a binary search tree? Suppose the numbers 17, 16, 12, 19, 14, 15, 11, 20, 27, 13 are inserted in that order into an initially empty binary search tree. The standard ordering for the given numbers is used by the binary search tree. Make a binary search tree and find the pre-order, in-order and postorder traversal sequence of the resultant tree? (10)

b) In the above tree, first 12 is deleted and after that 20 is deleted from the BST. Draw the BST after each insertion and deletion step, and perform pre-order, in-order, and post-order traversal of the resultant binary search tree. (10)

Q.5 (a) What is a graph? Discuss the methods of graph representation in memory with the help of suitable examples. (10)

(b) What is the minimum spanning tree? Solve the below graph using kruskal's algorithm. (10)

